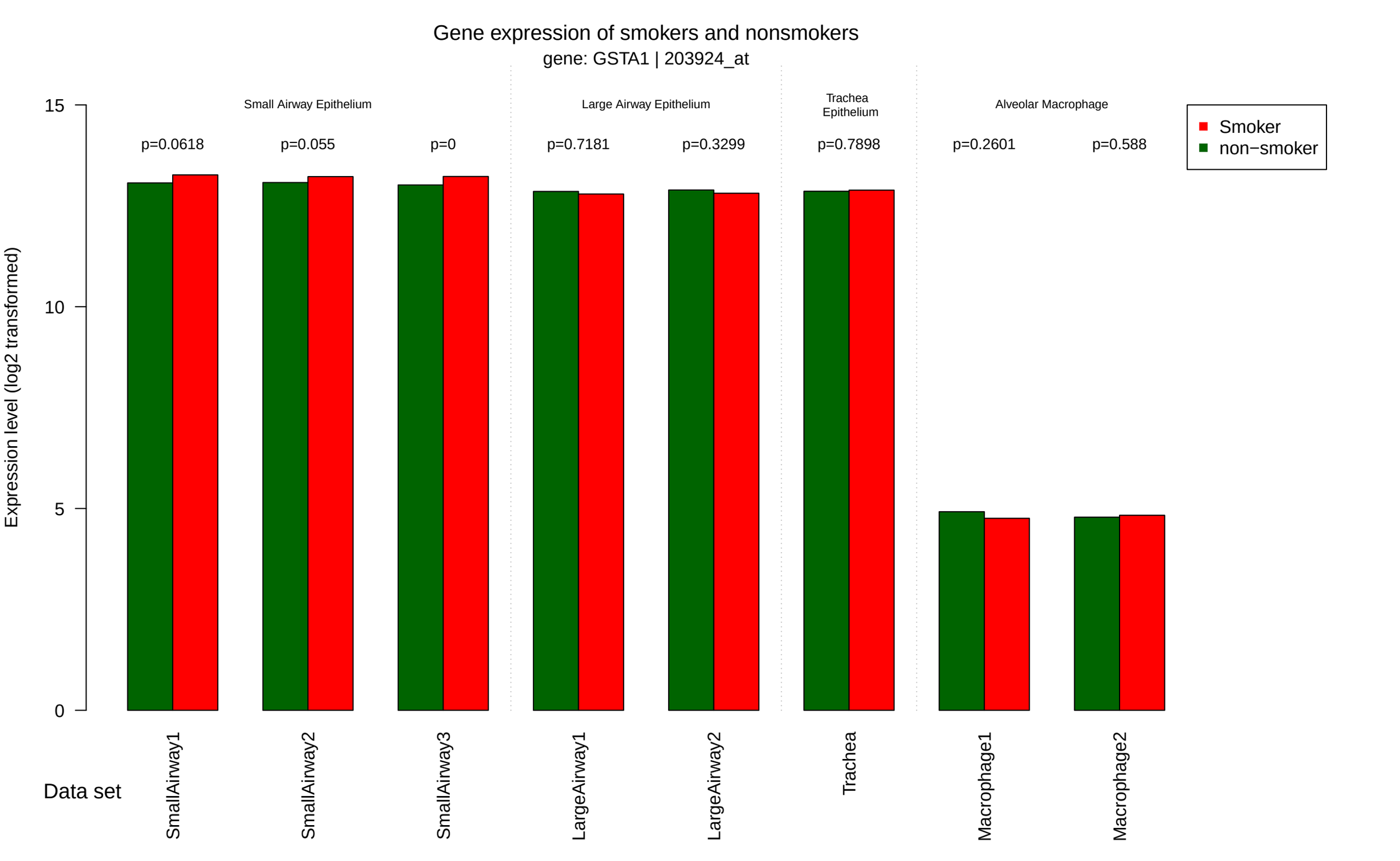
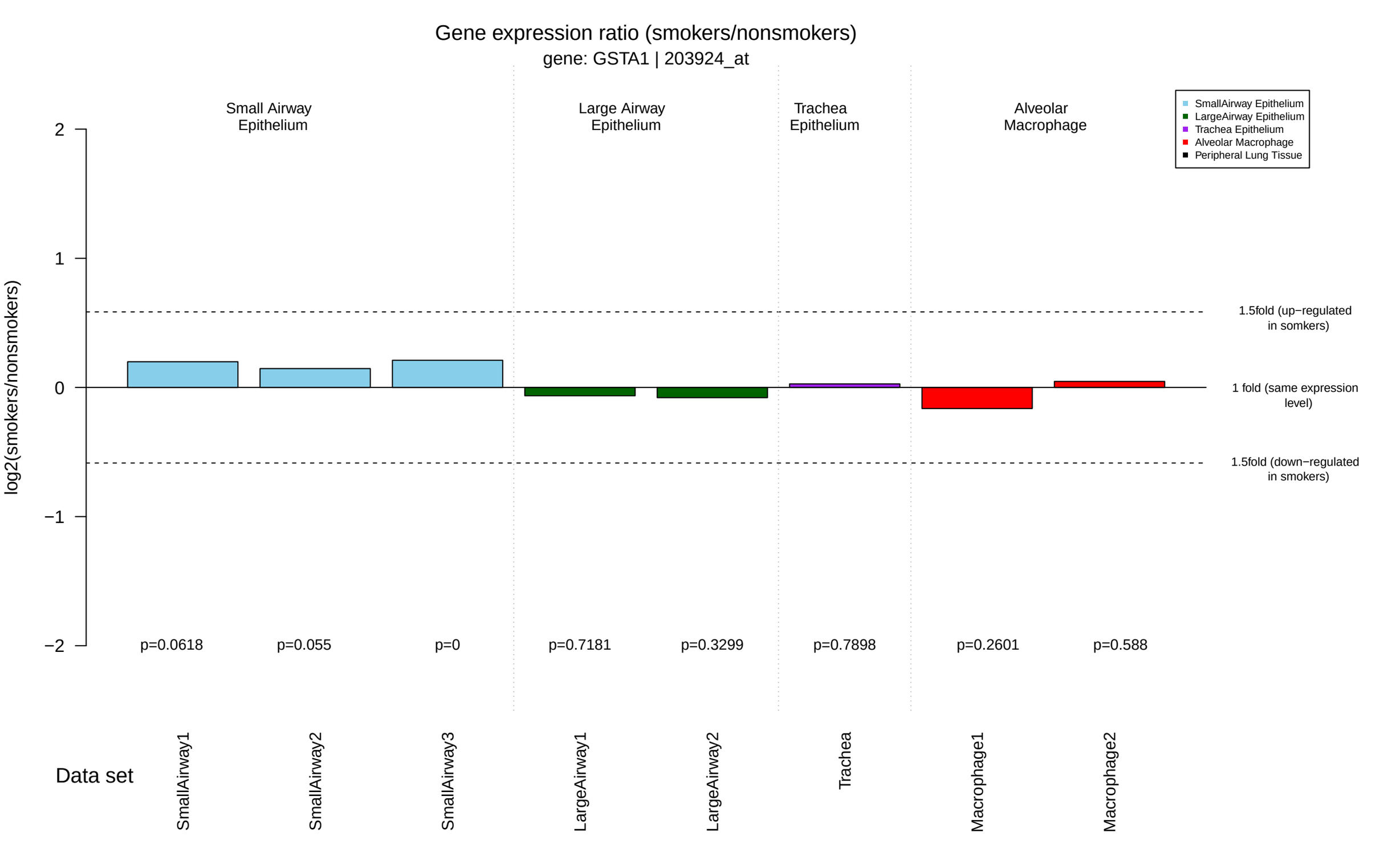
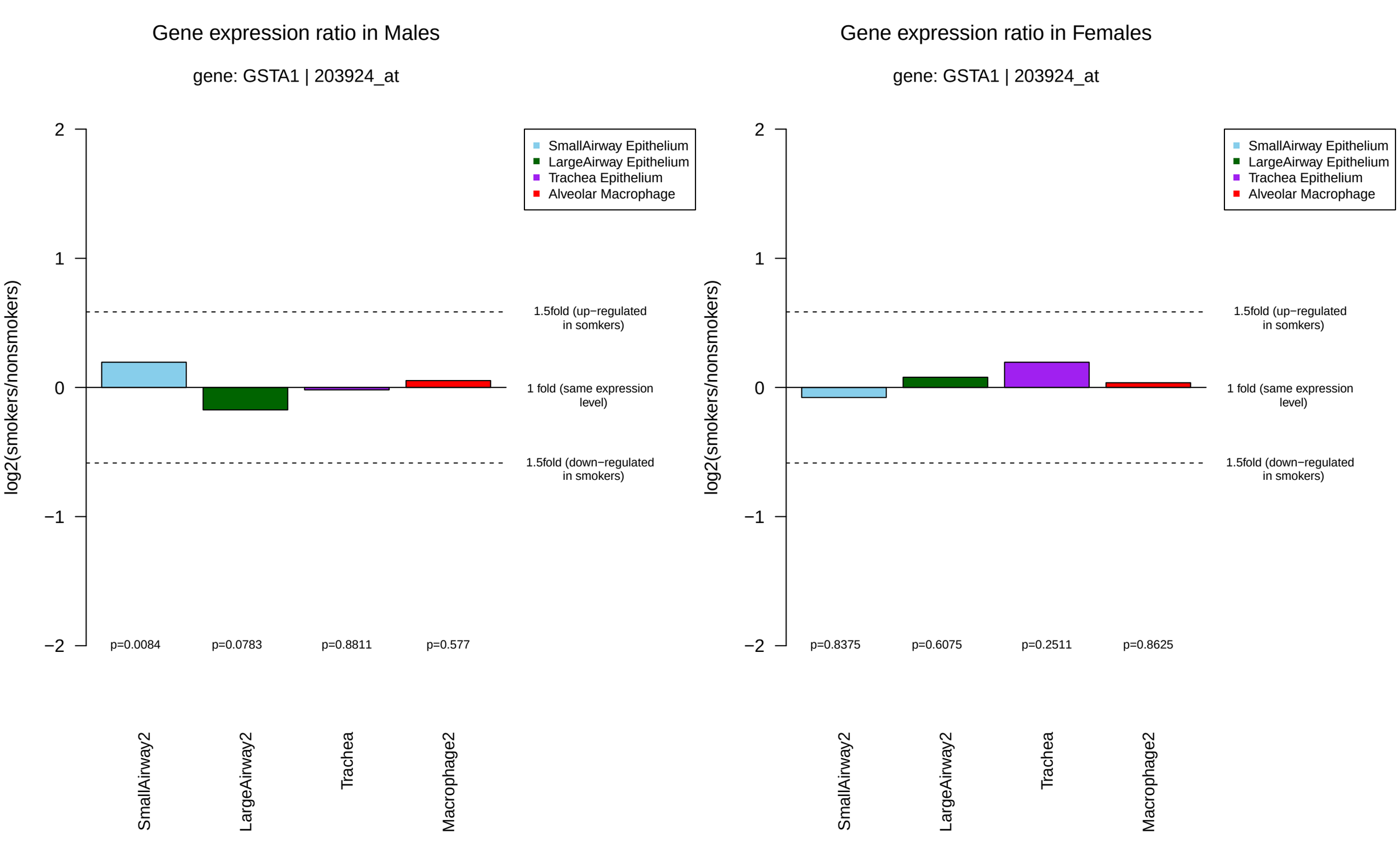
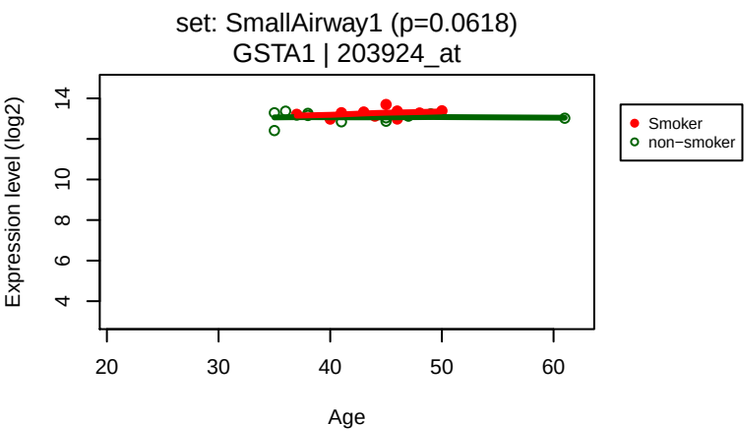
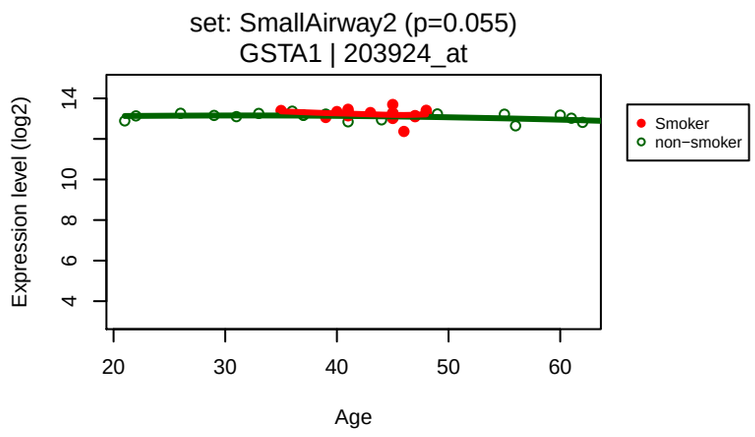
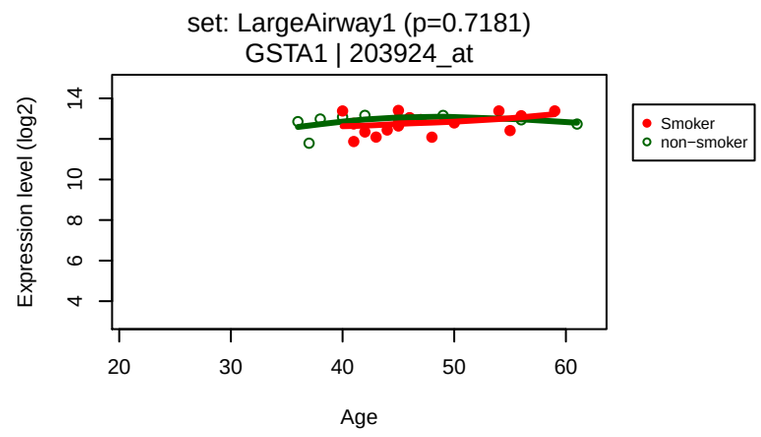

Gene expression of smokers and nonsmokers
gene: GSTA1 | 203924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0618
p=0.055
p=0
p=0.7181
p=0.3299
p=0.7898
p=0.2601
p=0.588
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

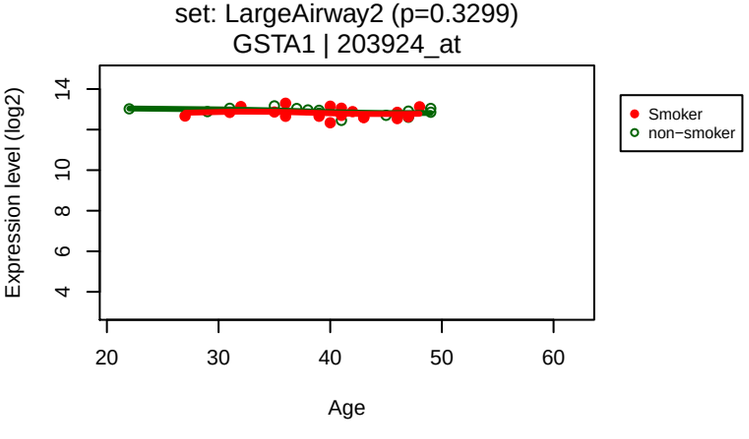

Gene expression ratio (smokers/nonsmokers)
gene: GSTA1 | 203924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0618
p=0.055
p=0
p=0.7181
p=0.3299
p=0.7898
p=0.2601
p=0.588
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

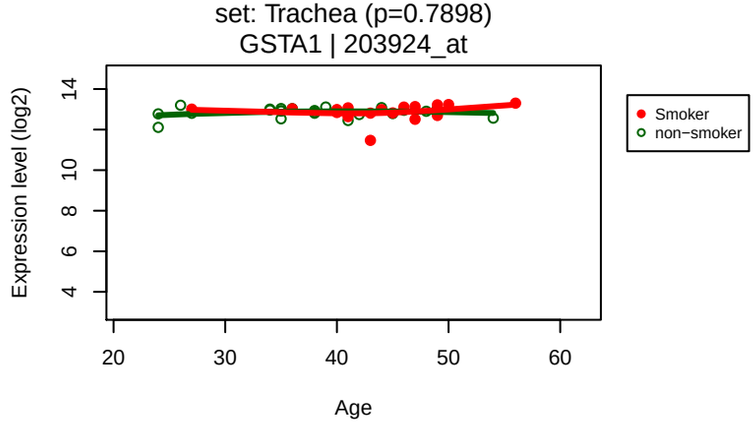

Gene expression ratio in Males
gene: GSTA1 | 203924_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0084
p=0.0783
p=0.8811
p=0.577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTA1 | 203924_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8375
p=0.6075
p=0.2511
p=0.8625
SmallAirway2
LargeAirway2
Trachea
Macrophage2

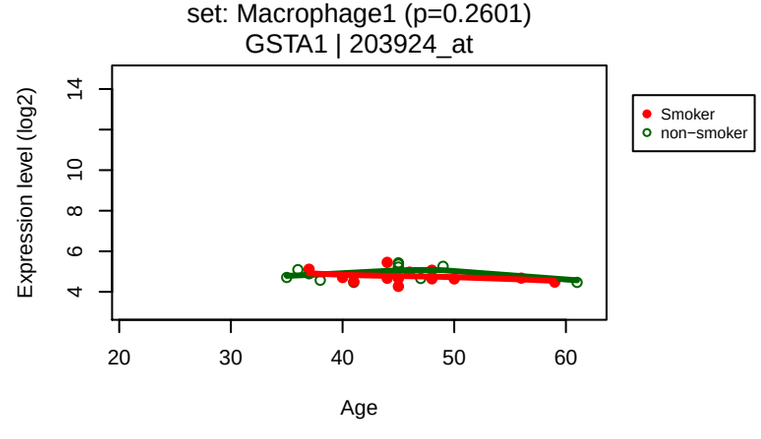

set: SmallAirway1 (p=0.0618)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker

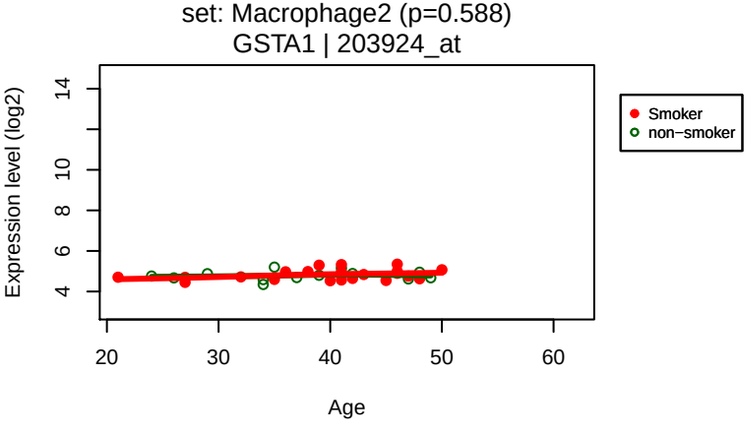

set: SmallAirway2 (p=0.055)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker

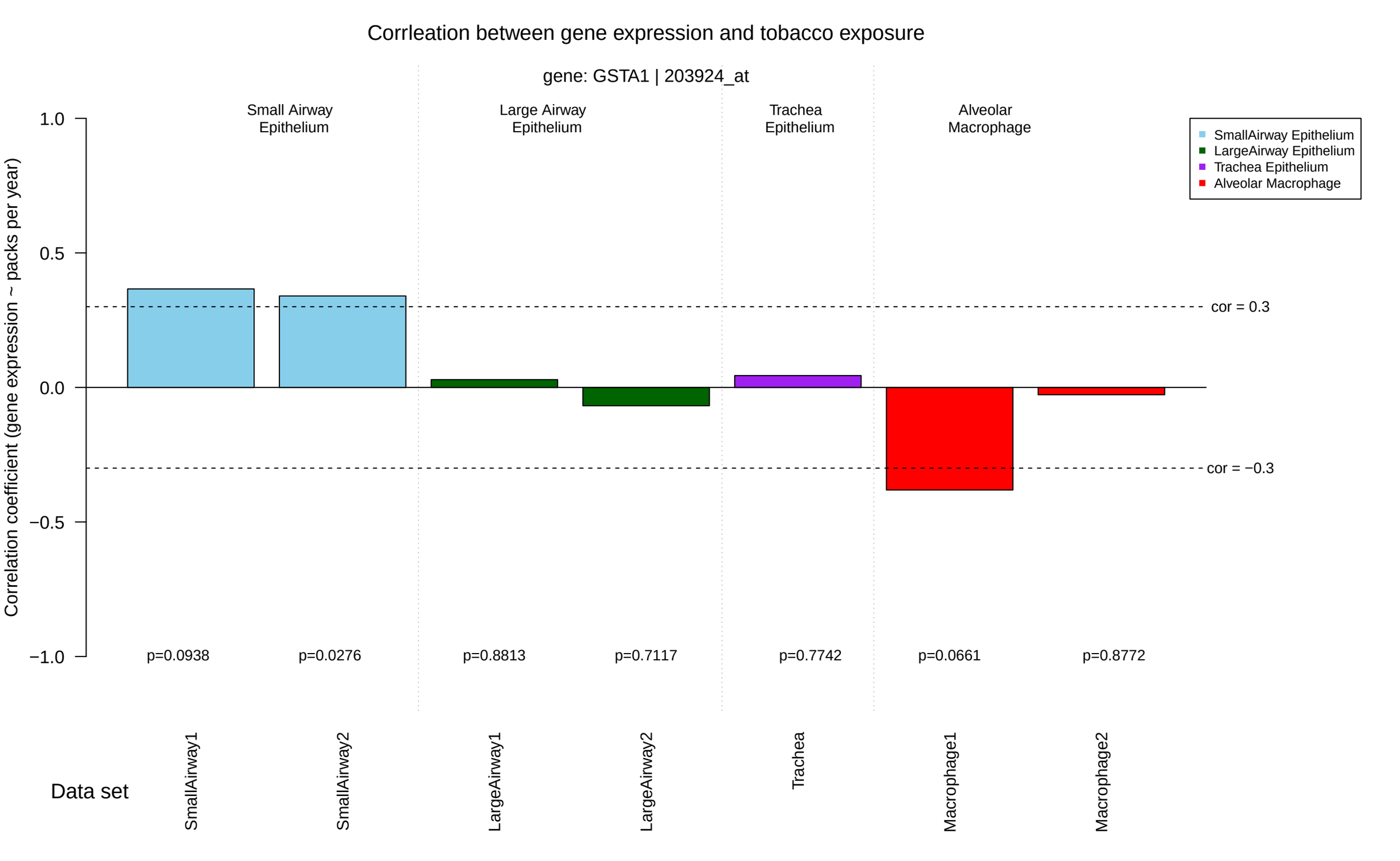

set: LargeAirway1 (p=0.7181)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3299)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7898)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2601)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.588)
GSTA1 | 203924_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTA1 | 203924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0938
p=0.0276
p=0.8813
p=0.7117
p=0.7742
p=0.0661
p=0.8772
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2